Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	7 February 2026
Team ID	LTVIP2026TMIDS86321
Project Name	**IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro**
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	System Setup	USN-1	As a developer, I can set up the development environment for the IntelliSQL project.	2	High	Team
Sprint-1	Database Design	USN-2	As a developer, I can create and configure the SQLite database to store student data.	2	High	Team
Sprint-1	Query Processing	USN-3	As a user, I can enter a natural language query through the web interface.	3	High	Team
Sprint-1	Query Processing	USN-4	As a system, I can process the natural language query and prepare it for SQL generation.	3	Medium	Team
Sprint-1	SQL Generation	USN-5	As a system, I can convert natural language queries into SQL queries.	3	High	Team
Sprint-2	User Interface	USN-6	As a user, I can access a simple Streamlit web interface to interact with the system.	2	High	Team

Sprint-2	**Query Execution**	**USN-7**	**As a system, I can execute the generated SQL query on the database.**	**3**	**High**	**Team**
Sprint-2	**Result Display**	**USN-8**	**As a user, I can view query results in a clear tabular format.**	**2**	**High**	**Team**
Sprint-2	**Data Analysis**	**USN-9**	**As a user, I can retrieve highest, lowest, or average marks from the database.**	**3**	**Medium**	**Team**
Sprint-2	**Filtering**	**USN-10**	**As a user, I can filter student data based on company placements such as Infosys or TCS.**	**3**	**Medium**	**Team**
Sprint-2	**Deployment**	**USN-11**	**As a developer, I can deploy the IntelliSQL application on Streamlit Cloud.**	**4**	**High**	**Team**

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	**13**	**6 Days**	**01 Feb 2026**	**06 Feb 2026**	**13**	**06 Feb 2026**
Sprint-2	**22**	**6 Days**	**08 Feb 2026**	**13 Feb 2026**	**22**	**13 Feb 2026**
Sprint-3	**0**	**6 Days**	**15 Feb 2026**	**20 Feb 2026**	**0**	**20 Feb 2026**
Sprint-4	**0**	**6 Days**	**22 Feb 2026**	**27 Feb 2026**	**0**	**27 Feb 2026**

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Velocity Calculation

Total Story Points = 35

Number of Sprints = 2

Velocity =

$$\boldsymbol{Velocity = \frac{Total\ Story\ Points}{Number\ of\ Sprints}}$$

$$\boldsymbol{Velocity = \frac{35}{2}}$$

$$\boldsymbol{Velocity = 17.5 \approx 18}$$

Team Velocity ≈ 18 Story Points per Sprint

Average Velocity (AV)

Sprint Duration = 6 days

Velocity = 18

$$AV = \frac{Sprint\ Duration}{Velocity}$$

$$AV = \frac{6}{18}$$

$$AV = 0.33$$

Average Velocity ≈ 0.33 story points per day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts